KERA
Kerala Climate Resilient Agri-Value Chain Modernization
IRRI
CGIAR

KERA-AWD Connect

Insights from research and field actions from **Kerala's** low-emission rice systems

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Navigating adversities: When every drop counts

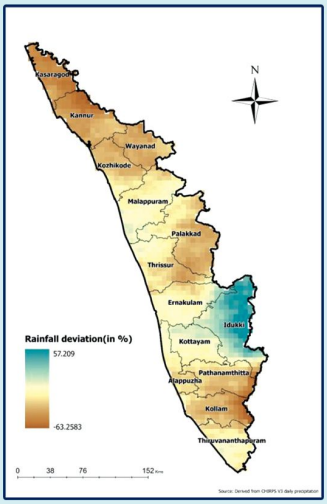


Fig. 01. Kerala Rainfall Deviation (1991- 2020)

Far away, an El Niño event is altering global weather patterns. Here in Keralam, its effects are unfolding through delayed and lesser rains, and lower inflows into reservoirs and canals. For a rice farmer, this is not just a change in the weather. It is a question of whether there will be enough water, at the right time, for the crop to grow. And to understand, from the real world, what happens when water becomes scarce or when fields may no longer remain continuously flooded, as they are usually kept?

To assess the early performance of the Southwest Monsoon in Keralam, we analysed CHIRPS v3 rainfall data for June 2026, which integrates satellite-derived rainfall estimates with observations from ground rain gauges. Daily rainfall totals for 1–30 June 2026 were compared with the long-term June average (1991-2020). The analysis indicated a weaker-than-normal start to the monsoon: statewide data assessment indicates an overall rainfall shortfall of approximately 15%, showing a clearer deficit. Palakkad received approximately 22.6% less rainfall than its thirty-year June average, placing it in the deficient category while the neighbouring Thrissur recorded a lesser deficit of 15.8%.

These are precisely the challenges that lie at the heart of the KERA-AWD initiative: understanding water availability better, using every drop more wisely, and identifying pathways to produce rice with lower water use and fewer greenhouse gas emissions (GHG) in an increasingly uncertain climate. From the flooded paddy tracts to research laboratories, and policy desks, this is the journey of state, scientists, scholars, extension officials, and farmers coming together to reimagine the future of rice cultivation.

Strengthening the Evidence Base

1.Advanced Crop Analytics and Data Projections

During last quarter, significant progress was made in strengthening the project's geospatial monitoring framework through the integration of satellite remote sensing and drone-based mapping. These activities generated high-resolution spatial datasets and crop monitoring products that enhance the understanding of rice distribution, crop growth dynamics, and landscape hydrology, providing a robust scientific foundation for precision water management and landscape-scale decision-making.

District-wide Rice Mapping and Phenology: Utilizing Sentinel-1 Synthetic Aperture Radar (SAR) and Sentinel-2 optical data highly precise, district-wide baseline rice area maps were created for Palakkad and Thrissur. By blending data from Sentinel-2, Landsat-8/9, and MODIS imagery, the team-built crop phenology analytics for *Mundakan* rice. Turning satellite time-series data into clear biological markers allows the team to map exactly when a crop starts, peaks, and ends its season.

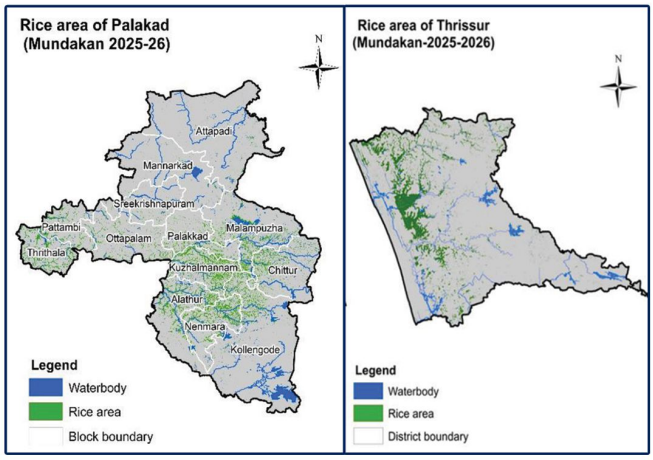


Fig. 02. Satellite-based rice area maps of (a)Palakkad and (b)Thrissur districts

Micro-topography and Drainage Models:

Post-harvest drone surveys across the Thrissur Kole lands generated high-resolution orthomosaics, Digital Surface Models (DSM), and Digital Terrain Models (DTM). These maps capture the micro-topography of the landscape, revealing subtle variations in bund height, field slopes, and drainage patterns. This detailed spatial information will support more precise and efficient water management across the landscape.

2.Development of Digital Soil Map

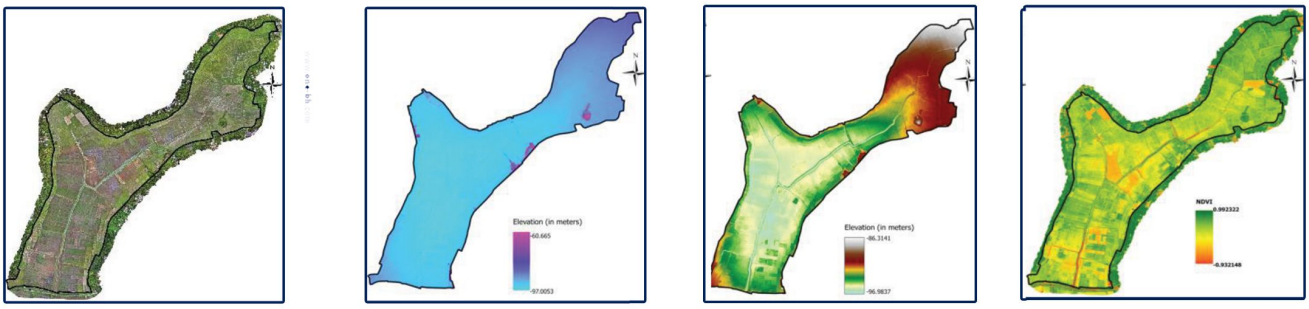


Fig. 03. a. RGB b. DSM c. DTM d. NDVI of Chathankole padasekharam

To construct highly accurate digital soil maps, the team deployed a robust, stratified random sampling framework factoring in land-use changes and local agro-ecological conditions. Field teams have been moving steadily across landscapes to collect vital samples:

- Palakkad: Samples collected from 112 locations
- Thrissur: Samples collected from 95 locations

With the remaining sites scheduled to be wrapped up by mid-July, the project is on track to deliver high-resolution digital soil maps to guide precision nutrient and soil management for years to come. Concurrently, multi-temporal satellite imagery is providing real-time soil moisture stress maps, allowing us to spot water constraints before they impact crop health.



Fig. 04. Soil Sampling in Puthukode block, Palakkad



Fig. 05. Soil Sampling in Chowannur block, Thrissur

3.Season 3 Station Trials: Exploring New Pathways

During the last two *Mundakan* seasons (2024-25 & 2025-26), the on-station experiments at ARS, Mannuthy, and RARS, Pattambi focused on evaluating different irrigation regimes (continuous flooding & Alternate Wetting and Drying (AWD), in combination with soil amendments, rice cultivars, and microbiome-based interventions to develop low-emission rice production packages. These studies generated valuable insights into strategies for reducing GHG emissions while maintaining crop productivity. Taking the research forward, the *Virippu*

season experiment aims to investigate the combined effects of different crop establishment methods (farmer broadcasting, drum seeding, and transplanting) and nitrogen management practices, including conventional urea broadcasting and urea deep placement (UDP), upon productivity, emissions, nutrient and water-use efficiency. Different crop establishment methods alter soil moisture regimes, root development, and field water requirements, thereby influencing irrigation demand, crop productivity, and GHG emissions. Nitrogen use efficiency (NUE) in rice is typically low, with only 30–40% of the applied nitrogen being utilized by the crop, while the remainder is lost through volatilization, leaching and runoff. These experiments aim to identify crop establishment and nutrient management practices that maximise resource-use efficiency, while simultaneously evaluating their effects on nitrous oxide (N_2O) emissions. The findings will contribute to the development of climate-smart, low-emission, and resource-efficient rice production systems.



Fig. 06. Sowing using seed drum technique in RARS Pattambi

4.Scaling up Research in Farmers' Fields



Fig. 07. GHG measurement using TGA in fallow fields

Crop, water and emission dynamics in paddy fields: On-farm experiments are currently underway for the *Virippu* 2026 season in six sluice based experimental field units in Palakkad. Experimental and control plots in selected padasekarams were mapped out, and soil amendments specifically biochar/phosphogypsum + lime were carefully applied. To understand the shifting water dynamics beneath the surface, field staff are now measuring soil moisture at strict four-day intervals. The first round of measurements to quantify nitrous oxide (N2O) and methane (CH4) fluxes has been completed in Palakkad using a Trace Gas Analyzer (TGA).

Water and emission dynamics in fallow fields: Following the *Mundakan* season, rice fields remain fallow for approximately 2–3 months in Palakkad and 4–5 months in Thrissur. GHG emissions were measured under varying crop residue levels and soil moisture conditions during this fallow period to improve our understanding of emission dynamics across the contrasting rice ecosystems of the two districts.



Fig. 08. GHG measurement in cropped land using TGA in Palakkad

Block	Panchayath	Experimental field	Control field
Alathur	Erimayur	Kannampulli	Kannampulli
Kuzhalmannam	Kannadi	Uppumpadam	Uppumpadam +
Malampuzha	Malampuzha	Vakkad	Thooppalam

Water and emission dynamics in ponded fallow fields: Monitoring water and emission dynamics in ponded fallow fields provides critical insights into off-season methane generation and water retention patterns. These observations help identify opportunities to reduce emissions without compromising future crop productivity. In the unique, water-logged ecosystem of the flooded Kole lands, standard measurement methods wouldn't suffice. The team developed and deployed specialized floating chamber-based TGA setups to capture gas fluxes drifting off the water surface. To continuously track standing water depth throughout the upcoming crop period, water level scales were fixed nearby.

Extensive monitoring: While field staff manage day-to-day operations, project associates and scientists conduct weekly inspections to guarantee research quality and consistency.



Fig. 09. GHG measurement in ponded fallow fields of Kole lands

5. Tracing the Emissions Footprint

The Shimadzu GC-2030 gas chromatograph was successfully installed and commissioned at ARS Mannuthy in March 2026, significantly enhancing the capacity to process large volumes of air and soil samples with speed and precision. To date, approximately 7,326 samples from Experiments 1 and 2 at ARS Mannuthy and ARS Pattambi have been successfully analysed. The team is now processing the next batch of 3,600 samples from Experiment 3, with completion targeted for August 2026. Meanwhile, modelling works are also progressing steadily. With the generation of 60,928 soil input files and the continued processing of historical weather datasets, the development of regional soil and weather databases for ORYZA modeling is advancing. First-season experimental, field survey, and GHG datasets have been compiled, standardized, and reviewed to support data cleaning, quality assurance, model calibration and validation, and subsequent scenario analyses. A standardized sampling and incubation protocol for estimating soil CH_4 production potential was developed to improve methane simulations and model parameterization, while ORYZA model calibration and validation activities advanced through the preparation of model input files and evaluation of the available experimental datasets. Regional modeling workflows were further advanced through the development of standardized data collection protocols, enhancement of data templates, regional assessment tools, and technical documentation to support large-scale GHG quantification.

Connecting Stakeholders, Shaping the Season

1. Conversations Before the Crop: Pre-Season Planning Workshop (*Virippu* 2026)

The workshop brought together scientists, government officials, extension personnel, and farmers to review progress, reflect on lessons from previous seasons, and prepare for the upcoming cropping season. Through technical sessions, farmer feedback, and collaborative discussions, participants reflected on field experiments, geospatial monitoring, irrigation strategies, modelling activities, and payment mechanisms to strengthen climate-smart, low-emission rice production. The workshop also facilitated interdisciplinary knowledge exchange through presentations by project researchers and Ph.D. scholars, fostering collaboration across institutions and work packages. The deliberations culminated in a coordinated roadmap for the upcoming season, emphasizing scientific rigour, stronger field implementation, enhanced data integration, and continued stakeholder engagement. The workshop also featured the official release of the Malayalam edition of the Keralam AWD Connect newsletter (Issue 2), ensuring that knowledge of AWD technology translates directly into the local language.


Fig. 10. Pre-seasonal planning workshop at KAU, Vellanikkara


Fig . 11. Group discussions after PhD presentation

2. Learning from Farmers: Feedback Sessions

To capture farmers' experiences and field-level insights from the first season of on-farm trials (*mundakan* 2025-26), a series of targeted feedback workshops were conducted across both experimental and control padasekharams. Nine workshops-four in Thrissur and five in Palakkad-brought together nearly 200 farmers to share their experiences, challenges, and recommendations. Each programme witnessed the active participation of Agricultural Department officials, Local Self Government (LSGD) representatives, KERA officials, and scientists, fostering constructive dialogue between farmers and technical stakeholders. Qualitative feedback methodologies, such as the dot-matrix ranking technique and problem-solution matrix, were utilized to capture farmer perspectives. The dot-matrix allowed every individual farmer to prioritize their observations regarding AWD outcomes, implementation issues, and future lines of action. This participatory problem analysis helped farmers suggest practical solutions and identify specific stakeholder roles required for implementation.

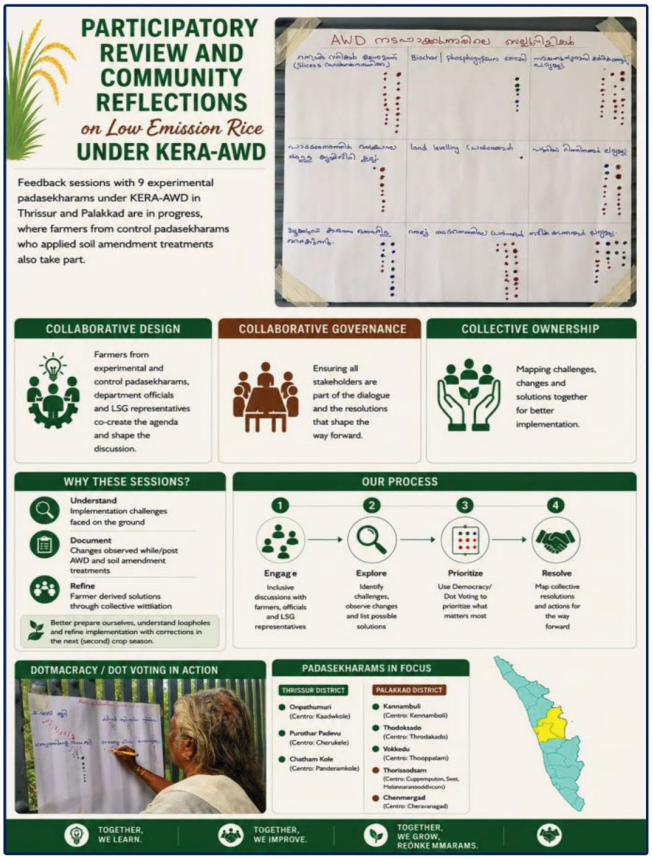


Fig. 12. Infographic brief on feedback workshops

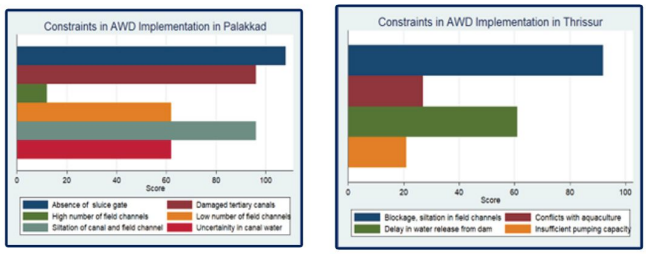


Fig. 13. Hydrological constraints to AWD implementation

3. Strengthening Community Readiness: Padasekharam Sensitization (*Virippu* 2026)



Fig. 14. Sensitization programme for Uppumpadam padasekharam

Sensitization activities were carried out across six sluice-based padasekharam units in Palakkad participating in the *virippu* 2026 KERA-AWD trials. The programmes reinforced farmers' understanding of key low-emission rice management practices, including AWD, the role of soil amendments in improving soil health and reducing GHG emissions, and the importance of coordinated water management for successful implementation. Such continued engagement is essential for strengthening farmers' confidence, addressing field-level concerns, and fostering the long-term adoption of low-emission rice cultivation practices.

Building Capacity, Growing Futures

1. Training Workshop on ORYZA MRV and Low-Emission Rice Systems



Fig. 15. Workshop on ORYZA- MRV

The ORYZA MRV Training Workshop on Low-Emission Rice Systems, conducted at Kerala Agricultural University from 27–30 April 2026 under the KERA-AWD Project, brought together 46 researchers, faculty members, technical staff, and

students representing universities, research institutions, government agencies, and the private sector. Led by the IRRI ORYZA Team, the four-day programme aimed to strengthen participants' capacity in the application of the ORYZA crop model for monitoring, reporting, and verification (MRV), GHG assessment, and low-emission rice systems research.

2. Internship for Kerala Agricultural University (KAU) - Experiential Learning Programme (ELP) students

The International Rice Research Institute (IRRI) in collaboration with the International Wheat and Maize Improvement Centre (CIMMYT) and the Foundation for Agrarian Studies (FAS) organised the internship programme on "Agrarian Studies and Social Science Research Methods" in collaboration with KAU ELP. The five-week residential programme brought together seven final-year undergraduate students from four Colleges of Agriculture under KAU—three from CoA Vellanikkara, two from CoA Vellayani, and one each from CoA Padannakkad and CoA Ambalavayal. Combining classroom learning, hands-on data analysis, field exposure, and mentorship from national and international experts, the programme strengthened participants' skills in research methodology, scientific writing, statistical analysis, and evidence-based decision-making. The internship culminated in a research symposium where students presented independent studies on diverse socio-economic and agricultural issues, including women's empowerment, fertilizer overuse, adoption of AWD, participation in voluntary carbon markets, climate-smart technology adoption, and the influence of social inequalities on digital agriculture, demonstrating the programme's commitment to nurturing the next generation of agricultural researchers.

Beyond Books: My lasting internship experience at ICRISAT

"The five-week internship under the IRRI/CIMMYT/FAS/KAU programme at ICRISAT, Hyderabad, was a truly enriching learning experience. It strengthened our research skills through hands-on training in R programming, STATA, econometrics, regression analysis, Kobo Toolbox, remote sensing, and carbon credit economics, guided by experienced resource persons. A major highlight was the field visit to Nizamabad, where we observed direct seeded rice (DSR) cultivation and laser land leveller technology, and interacted with lead farmers, connecting classroom concepts with real-world agricultural practices. Interactions with experts from organizations such as UNDP, NIAP, NITI Aayog, Mahyco, and ILRI further broadened our understanding of how research contributes to agricultural policy, innovation, and sustainable development. Overall, the internship was an inspiring journey that enhanced both our technical knowledge and confidence as young researchers".

- Mohammed Fayiz MP, B.Sc. (Hons.) Agriculture, CoA Vellanikkara, KAU



Fig. 16. The internship cohort together during the five-week programme at ICRISAT, Hyderabad



Fig. 17. Students interacting with the farmer coordinators during the field exposure visit to Nizamabad district

Voices that Matter



Mr Satheeshkumar M D
(Deputy Director of Agriculture (WM), Palakkad)

Beyond AWD experiments, our Ground Truthing Survey in Palakkad is creating a powerful database for tomorrow. Through LULC Mapping, Paddy area validation, crop delineation, MRV and Geospatial monitoring, we are laying the groundwork for smarter, more efficient use of our resources

AWD is not just a water-saving technique; it is a climate-smart approach that helps farmers produce more with less water while protecting our environment for future generations. This method helps save 20–30% of irrigation water without reducing crop yield. It also lowers methane emissions from paddy fields, reduces irrigation costs, improves water-use efficiency, and supports sustainable management of precious water resources. By using water judiciously, farmers can cultivate paddy more economically while contributing to climate change mitigation



Ms Sherinlincy A
(Agricultural Officer, Erimayur Panchayat)



Mr K T Prasad
(Executive committe member, Kannampulli padasekharam, Palakkad)

AWD has great potential to save water and make rice farming more sustainable. But unless our canals and sluices are repaired, water losses and leakages will continue to undermine its success. We hope the repair works planned under KERA will prioritise these critical irrigation structures and make effective water management a reality.

Meet the Team behind KERA-AWD

(A recurring series introducing the people behind the project)

Dr Anton Urfels



Dr. Anton is a Senior Scientist in Water Management at the International Rice Research Institute (IRRI), where he leads research on agricultural water management, climate adaptation, and sustainable intensification of rice-based farming systems. He co-leads KERA-AWD from IRRI.

He holds a Ph.D. in Agricultural and Environmental Science from Wageningen University and Research, the Netherlands, specializing in crop management, irrigation systems, and farmer decision-making in South Asia. Prior to IRRI, he worked with CIMMYT in Nepal as a Systems Agronomist and Agricultural Water Systems Consultant, focusing on climate resilience, groundwater management, and irrigation modernization. Dr. Urfels has published extensively on irrigation development, groundwater governance, and climate adaptation. He is a Visiting Fellow at Cornell University and Wageningen University, and his work spans South and Southeast Asia, supporting governments and farming communities on food security, water scarcity, and climate change

Dr Thulasi V



Dr. Thulasi holds both her Master's and Doctoral degrees in Soil Science and Agricultural Chemistry from the Indian Agricultural Research Institute (IARI), New Delhi. As a Scientist under the KERA-AWD project, she leads station-based experiments at RARS Pattambi and supports the field-scale implementation of low-emission agronomic packages tailored for Kerala's paddy ecosystems. At the Regional Agricultural Research Station (RARS), Pattambi, Dr. Thulasi has played a pivotal role in developing innovative multi-nutrient formulations, including the KAU Sampoorna Multimix, and formulating recommendations for sustainable nutrient management and carbon sequestration in rice-based systems. Her contributions to agricultural research have earned her more than 20 national and state-level awards, including the IARI Merit Medal (2006), S.P. Rai Chaudhury Gold Medal, Young Scientist Award at the Kerala Environment Congress (2012), and the Best Researcher Award from the Department of Agriculture, Government of Keralam (2022). Her research interests include soil fertility, soil biology, carbon sequestration, crop simulation modelling, and the development of climate change mitigation strategies, with a particular focus on advancing sustainable and low-emission agricultural systems.

Dr Chinni V Naga Kumar Kommireddi

Dr. Chinni obtained his PhD in Geo-Engineering from Andhra University, Visakhapatnam. He also earned his M.Tech. degree in Remote Sensing and GIS, jointly awarded by Andhra University and the Centre for Space Science and Technology Education in Asia and the Pacific (affiliated to United Nations) IIRS Campus, Dehradun. He is currently serving as a Senior Scientist & Head, Land and Water Management Research Group at the Centre for Water Resources Development and Management (CWRDM), Kozhikode, Keralam. Dr. Chinni has made significant contributions to the applications of geospatial technologies in the fields of coastal evolution and morphodynamics, natural disasters (with emphasis on cyclones, floods, geomorphological aspects of land subsidence), and water resources management (particularly in the mapping of paleochannels and paleo-environment reconstruction).



Mr Jacob Joy



Mr Jacob is currently serving as Deputy Director of Agriculture and Technical Officer, KERA-SPMU. He holds a Postgraduate Degree in Agricultural Meteorology from Punjab Agricultural University and earned his bachelor's degree in agriculture from Kerala Agricultural University. With over 23 years of service in the Department of Agriculture, he has acquired extensive experience across a wide range of technical and administrative functions. He has also completed a Postgraduate Diploma in Plant Health Management from National Institute of Plant Health Management and a Postgraduate Diploma in ICT and Social Media from National Institute of Electronics and Information Technology. During his career, he has served in various capacities, including administrative, marketing, and farm management roles, across several districts of Keralam, including Thiruvananthapuram, Malappuram, and Wayanad. His broad professional experience, combined with his expertise in agricultural meteorology, plant health management, and information and communi cation technology, has contributed significantly to the implementation of agricultural development initiatives of the Department, including the KERA Project.

Mr Joseph John Therattil



Mr Joseph holds a B.Sc. in Agriculture from Kerala Agricultural University, a Post Graduate Diploma in Plant Health Management from the National Institute of Plant Health Management (NIPHM), Hyderabad, and a master's degree in journalism and mass communication from Bharathiar University and MBA from IGNOU. During his distinguished career spanning over 26 years, he has received the State Best Farm Journalist Award (Karshaka Bharathi) and the State Best Agricultural Officer Award from the Government of Kerala. He was adjudged as a State Resource Person in Agriculture and management. His professional contributions include conducting more than 750 farmer training programmes and publishing over 600 technical and popular articles, and authoring three books on vegetable cultivation and sustainable agriculture. He is presently serving as Deputy Regional Project Director of the World Bank-assisted KERA Project, Government of Kerala, where he leads the implementation of climate-resilient agriculture, value chain modernization, agribusiness promotion, Farmer Producer Organizations (FPOs), Agri-Tech start-ups, and commercialization initiatives across the districts of Ernakulam, Thrissur, Palakkad, and Malappuram.

Dr Syama S Menon



Dr. Syama is an Assistant Professor cum Technical Specialist (Agronomy) and heads the Project Implementation Unit, Kerala Agricultural University, under the KERA, based at the Agricultural Research Station, Mannuthy. In this role, she coordinates all activities undertaken by Kerala Agricultural University in collaboration with KERA.

She holds a Ph.D. in Agronomy from KAU, specializing in irrigation and stress management. She began her career as an Agricultural Officer in 2017 and later joined KAU as an Assistant Professor in 2019. Since 2014, she has published around 14 journal papers and has developed expertise in nutrient and water management. She has guided three postgraduate students and contributed to several state plan projects.

Ms Nithya Ajith



Ms. Nithya completed her Diploma in Agricultural Science from the Institute of Agricultural Technology (IAT), Regional Agricultural Research Station (RARS), Pattambi, and holds a B.Sc. in Physics. Although her academic journey began in the physical sciences, her passion for agriculture motivated her to pursue specialised training in agricultural science, where she developed strong theoretical knowledge and practical skills in crop production and sustainable farming practices.

As a Skilled Assistant with the KERA-AWD project, Nithya actively supports agricultural research, extension activities, and field-based implementation. Her enthusiasm for learning, combined with her multidisciplinary background and hands-on experience, enables her to make valuable contributions to research and farmer engagement activities.

Ms Amrutha N. S.



Amrutha N. S. holds a master's degree in environmental science from University of Calicut. As a Project Associate under the KERA project, she is responsible for UAV image processing, GIS-based landscape analysis, and the development of spatial databases and thematic maps to support project implementation and environmental monitoring. Prior to joining KERA, she served as a Research Assistant at the Kerala Forest Research Institute (KFRI), where she contributed to ecological restoration projects through the monitoring of biotic and abiotic parameters in eco-restored sites. Her professional interests include restoration ecology, soil science, ecosystem monitoring, and the application of GIS and remote sensing technologies for sustainable environmental management.



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